This is the start of the celebrated Baker-Campbell-Hausdorff series [21] (often referred to as the BCH Lemma for short) which gives an explicit formula for the product of exponentials of non-commuting operators.

Higher order terms can also be worked out, for example,

$$e^A e^B = \exp(A + B + \frac{1}{2}[A, B] + \frac{1}{12}([A, [A, B]] - [B, [A, B]]) - \frac{1}{24}[B, [A, [A, B]]] + \ldots)$$

and, using the correspondence between Lie and Poisson brackets,

$$e^{h\mathcal{L}_{H_1}} e^{h\mathcal{L}_{H_2}} = e^{h\mathcal{L}_{\tilde{H}_h}},$$

where

$$\tilde{H}_h = H_1 + H_2 + \frac{h}{2}\{H_1, H_2\} + \frac{h^2}{12}(\{H_1, \{H_1, H_2\}\} - \{H_2, \{H_1, H_2\}\}) - \frac{h^3}{24}\{H_2, \{H_1, \{H_1, H_2\}\}\} + \ldots$$

We refer to this series as the *modified* (also *perturbed*, *shadow*) Hamiltonian corresponding to the splitting method. The implication of the series is that the numerical method may be viewed as being equivalent to the exact solution of a nearby Hamiltonian system, although we have not addressed the convergence of the expansion.

If H_1 and H_2 Poisson-commute, i.e.

$$\{H_1, H_2\} = 0,$$

then there is no error in splitting. This is an extreme example, but there are a number of special cases of splittings such that some of the terms vanish from the expansion, for example where H_1 or H_2 is quadratic, or where there are restricted dependencies.

3.3.1 Symplectic Euler

Splitting our Hamiltonian using $H_1 = T(\boldsymbol{p}) = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}/2$ and $H_2 = U(\boldsymbol{q})$ gives the symplectic Euler method. The BCH expansion gives the perturbed Hamiltonian

$$\tilde{H}_h = H - \frac{h}{2}\boldsymbol{p}^T \boldsymbol{M}^{-1} \nabla U(\boldsymbol{q}) + \frac{h^2}{12}\left(\boldsymbol{p}^T \boldsymbol{M}^{-1} U''(\boldsymbol{q}) \boldsymbol{M}^{-1} \boldsymbol{p} + \nabla U(\boldsymbol{q})^T \boldsymbol{M}^{-1} \nabla U(\boldsymbol{q})\right) - \frac{h^3}{12} \nabla U(\boldsymbol{q})^T \boldsymbol{M}^{-1} U''(\boldsymbol{q}) \boldsymbol{M}^{-1} \boldsymbol{p} + \mathcal{O}(h^4).$$